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Memristive Edge Compute-In-Memory Hardware: A Cross-Layer Exploration of Oxide-RRAM and Phase-Change Memory Architectures for Low-Power Vector-Matrix Multiplication

As edge computing applications continue to demand real-time data processing under tight energy budgets, traditional von Neumann architectures suffer severe performance degradation due to the memory-wall bottleneck. Compute-In-Memory (CIM) using memristive crossbar arrays offers a promising non-von Neumann alternative by executing physical Vector-Matrix Multiplication (VMM) directly within non-volatile memory arrays using Ohm's and Kirchhoff's circuit laws.

This theoretical and simulation-driven study presents a cross-layer evaluation comparing Hafnium Oxide-based Resistive Random-Access Memory (HfO_x-RRAM) and Phase-Change Memory (PCM) crossbar architectures for edge neural network acceleration. Through a systematic synthesis of device-level non-idealities, including conductance drift, cycle-to-cycle variability, and parasitic IR drop, this research models the trade-offs between device non-linearities and network inference accuracy on low-power edge benchmarks.

1. Research Questions & Hypotheses

Research Questions:

1. How do the physical device non-idealities of HfO_x-RRAM (e.g., stochastic filament switching) and PCM (e.g., structural relaxation resistance drift) quantitatively impact the computational accuracy of edge VMM crossbars?
2. What are the performance and energy trade-offs associated with peripheral mixed-signal interfaces (Analog-to-Digital Converters / Digital-to-Analog Converters) when scaling crossbar array sizes from 64×64 to 256×256?

3. To what extent can noise-aware mapping algorithms and differential 1T1R (One-Transistor One-Memristor) cell configurations mitigate the degradation of inference accuracy caused by sneak paths and parasitic IR drops?

Hypotheses

- Hypothesis 1: Incorporating noise-aware training models that inject measured cycle-to-cycle conductance variability into weight matrices will recover up to 90% of accuracy loss incurred by physical memristive non-idealities during edge inference.
- Hypothesis 2: While PCM architectures will demonstrate superior multi-level cell state stability for static weight storage, HfO_x-RRAM crossbars will yield higher overall energy efficiency (TOPS/W) due to lower programming pulse voltages and faster state transitions.

2. Literature Review

2.1 The Von Neumann Memory Wall and Compute-In-Memory Paradigm

In conventional computing systems, moving data repeatedly between separate processing units (CPUs/GPUs) and off-chip DRAM accounts for up to 80% of total system energy consumption [1]. Memristive hardware overcomes this physical separation by performing analog vector-matrix multiplication ($Y = X \cdot W$) directly within crossbar memory arrays [2], [3].

When input voltages (V_i) are applied to the rows of a memristive array, the current (I_j) collected along each column represents the instantaneous dot-product dictated by Ohm's Law and Kirchhoff's Current Law:

$$I_j = \sum_{i=1}^{N} V_i \cdot G_{ij}$$

where G_{ij} represents the programmable conductance state of the memristive device at row i and column j [2], [4].

2.2 Emerging Memristive Device Technologies

1. Oxide RRAM (HfO_x/TaO_x): Operates via the voltage-induced formation and rupture of nanoscale oxygen vacancy conductive filaments [3], [5]. RRAM offers high integration density, fast switching speeds (<10 ns), and native CMOS process compatibility [5]. However, oxygen vacancy migration introduces stochastic switching behaviour and cycle-to-cycle (C2C) conductance fluctuation [6].
2. Phase-Change Memory (PCM): Relies on thermal phase transitions between a disordered amorphous state (high resistance) and an ordered crystalline state (low resistance) driven by electrical pulses [7], [8]. PCM exhibits high endurance and multi-level cell capability, but suffers from structural relaxation over time, leading to resistance drift and accuracy degradation in deep networks [8], [9].

2.3 System-Level Challenges in Memristive Hardware

Despite hardware efficiency gains, deployable edge CIM accelerators face key non-idealities:

- Peripheral Circuit Overhead: Mixed-signal converters (ADCs and DACs) consume the vast majority of chip area (>60%) and energy budget (>70%) [10], [11].
- Parasitic Line Resistance (IR Drop): Metal interconnect resistance causes voltage drops along array rows and columns, leading to degraded computational precision in larger crossbar dimensions [12].
- Sneak Path Currents: Leakage current paths through unselected neighbouring cells degrade signal-to-noise ratios, requiring access-device isolation such as $1\mathrm{T1R}$ array architectures [13].

3. Methodology

3.1 Device Modeling and Dataset Preparation

Experimental memristance parameters (e.g., G_{max}, G_{min}, non-linearity factors, and conductance state quantization levels) will be extracted from peer-reviewed literature on 65 nm CMOS-integrated HfO_x-RRAM and PCM test chips [5], [8]. These physical device models will be integrated into the NeuroSim cross-layer simulation framework and MemTorch to evaluate hardware-level non-idealities.

3.2 Array & Circuit Simulation Framework

1. Array Configuration: Model a differential $1\mathrm{T1R}$ crossbar architecture to represent positive and negative neural weights ($W_{ij} \propto G_{ij}^{+} - G_{ij}^{-}$) [12].
2. Parasitic Modeling: Introduce finite metal wire resistance ($R_{\text{wire}} = 2.5\ \Omega$/cell) along array rows and columns to simulate non-linear IR voltage drop [12].
3. Peripheral Modeling: Simulate 1-bit to 8-bit DAC input resolution and 4-bit to 8-bit Successive Approximation Register (SAR) ADCs for column readout, calculating total energy consumption across array sizes (64×64, 128×128, and 256×256) [10], [11].

3.3 Algorithmic Mapping & Noise-Aware Compensation

Quantized Deep Neural Networks (Convolutional Neural Networks for CIFAR-10 classification and Keyword Spotting SNNs) will be trained using PyTorch. Weights will be mapped to discrete conductance levels. Noise-aware training will be applied by introducing Gaussian conductance noise during the forward pass:

$$G_{\text{actual}} = G_{\text{target}} \cdot \left(1 + \mathcal{N}(0, \sigma^2)\right)$$

where σ models physical device-to-device and cycle-to-cycle switching variability [6], [14].

4. Expected Outcomes

1. Quantified Hardware Benchmarks: Comprehensive evaluation of HfO_x-RRAM vs. PCM crossbars across computational density (TOPS/mm^2), energy efficiency (TOPS/W), and peak edge inference throughput.
2. Non-Ideality Impact Map: Detailed breakdown showing how parasitic IR drop, ADC bit-precision, and conductance drift independently degrade classification accuracy on edge workloads.
3. Optimized Mapping Framework: Verification that noise-aware algorithm training restores classification accuracy to within 1.5% of ideal floating-point software baselines on physical memristive crossbars.

5. Project Timeline & Practicalities

- Months 1–2: Comprehensive literature synthesis, extraction of memristive device conductance parameters, and setup of the NeuroSim / MemTorch simulation pipeline.
- Months 3–5: Implementation of 1T1R crossbar array models, inclusion of ADC/DAC energy models, and execution of noise-aware neural network training experiments.
- Months 6–7: Data extraction, statistical analysis of accuracy vs. energy efficiency trade-offs, and preparation of figures/visualizations.
- Month 8: Full report write-up, formatting, citation verification, and preparation of the final video pitch submission.

6. Conclusion

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This cross-layer study compares HfO_x-RRAM and PCM crossbar architectures to overcome the von Neumann memory bottleneck in low-power edge computing. By combining noise-aware training with differential 1T1R cell configurations, the proposed framework mitigates physical device non-idealities—such as parasitic IR drops, sneak paths, and conductance variability—recovering neural network inference accuracy to within 1.5% of software baselines. Ultimately, this research provides a quantitative roadmap for developing robust, energy-efficient Compute-In-Memory hardware tailored for real-time edge workloads

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